

Architectural Justification — Coffee Kiosk System

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (User Interface Component handling the touch-screen), a Business Layer (Order Manager, Payment Service, and Receipt Printer Components), and a Data Layer (Database Component storing menu and pricing data).

Reason 1 — Simplicity Matches the Scenario Scale

The kiosk supports a small, fixed feature set (3 coffee types, 2 sizes, single payment method) running on a single device. A Layered Architecture keeps this scope simple to build, test, and maintain, without the deployment and network overhead that Microservices would introduce for a system this small.

Reason 2 — Clear Separation for Hardware Integration

The kiosk must interact with two distinct hardware interfaces — the touch screen and the receipt printer. Layering isolates these hardware-facing concerns in the Business Layer behind well-defined interfaces, so the Presentation Layer and Database Layer are not directly coupled to hardware specifics and can be modified independently.

Security Advantage

By routing all credit card transactions exclusively through the Payment Service Component in the Business Layer, the Presentation Layer never directly touches payment data. This confines PCI-relevant logic and card-data handling to a single, well-isolated component rather than spreading it across the UI, reducing the system's attack surface.

Performance Benefit

Since all layers run within a single kiosk application rather than communicating over a network (as Microservices would require), there is no inter-service network latency. Order placement, payment confirmation, and receipt printing happen through direct in-process calls, keeping the customer-facing transaction fast.